# Texas Instruments (NASDAQ: TXN) — Company Research Dossier — Masterclass Companion Book

*A step-by-step narrative walkthrough and audio companion to Texas Instruments (NASDAQ: TXN) – Company Research Dossier: deconstructing the business engine, global historical parallels, ten-year financial forensics, both sides of the institutional debate, and the discipline of high-asymmetry execution.*

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*A Sequential Guide to the Business Engine, Global Parallels, Financial Forensics & High-Asymmetry Execution*


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SESSION PRICE

**$295.73**

VERDICT

**WAIT_FOR_SUPPORT**

STAGE POSTURE

**STAGE_2**

BUY / SUPPORT ZONE

**$280.15–296.95**

PRICE TARGET

**$318.60 / $334.03**

### PREFACE & CORE THESIS

Texas Instruments (TI) sells analog and embedded-processing chips for industrial, automotive, data-center and other electronic systems. Its broad catalog, long customer design lives and owned factories support durable demand, but chip cycles still affect results. The company is investing heavily in 300mm factories, so future cash returns depend on good use of that capacity. TXN closed at $295.73 on November 6, 2026, after a sharp recovery that left the shares at a premium valuation. Our view is WAIT_FOR_SUPPORT: require a fresh support retest and stronger cash-flow evidence before adding.

PROLOGUE

## The Business, The Price & Our Independent Verdict

> *"The investor's chief problem—and even his worst enemy—is likely to be himself."*
>
> — BENJAMIN GRAHAM

Texas Instruments sells foundational chips: analog devices that sense, condition, convert, and manage signals and power, plus embedded processors that control the equipment around them. Its customers build factory automation, vehicles, data centers, communications systems, and personal electronics. A customer may use one TI component in a board or many across a platform. The product is small; the qualification history, availability promise, engineering support, and manufacturing capability behind it are the value proposition. In 2025 TI generated $17.68 billion of revenue, of which Analog was $14.01 billion, Embedded Processing $2.70 billion, and Other $979 million. The company sells a broad catalog through direct accounts, distributors, and online tools, and invests heavily in internal wafer production.

At the November 6, 2026 close of $295.73, TXN had gained roughly 84% over the prior year and traded at 38.37 times trailing earnings and 36.93 times free cash flow on the live consumer snapshot. The rally reflects a sharp operating recovery: Q3 2026 revenue increased 28% year over year to $6.06 billion, and operating profit increased 73%. TI's management cited continued broad growth in industrial, automotive, and data center. The counterweight is that the stock is priced for more than a return to trough conditions. A 300mm capacity build remains capital-intensive, the adjusted free-cash-flow definition includes CHIPS incentives, and a planned Silicon Labs acquisition adds integration and financing requirements.

The bull thesis is that the catalog's longevity, direct customer relationships, and 300mm cost curve turn a cyclical recovery into durable free-cash-flow-per-share growth. The bear thesis is that a premium multiple meets fixed costs, uneven end demand, and capacity arriving ahead of demand. The business quality is high; the valuation cushion is not. We therefore separate three horizons: near-term price action is constructive but sector breadth remains weak;

medium-term fundamentals need another quarter or two of broad confirmation; long-term compounding remains plausible if the fab returns and acquisition economics prove sound. Our independent verdict is WAIT_FOR_SUPPORT, conviction tier TIER_1, strategy bucket CORE_ALPHA. The snapshot support band is $280.15–$296.95, and its invalidation reference is $273.39. These are monitoring levels, not orders or price targets.

For a short-term trader, do not chase a large advance; reassess after a support retest, a fresh technical snapshot, and confirmation from the wider semiconductor group. For a medium-term investor, watch bookings, revenue breadth, factory utilization, gross margin, inventory, and the cash bridge from operating flow through capex and incentives. For a long-term investor, TI belongs on a quality watchlist, but new capital should wait for either a better price or more evidence that cash generation per share can grow after investment. We assign no fundamental target because a complete, reconciled post-acquisition free-cash-flow model is not available in this dossier. A number without a credible depreciation, incentive, share-count, and financing bridge would be false precision.

Our decision is conditional and reviewable. We would become more constructive if TI holds support, reports continued end-market breadth, raises margins through loading rather than one-time items, and demonstrates cash growth that remains attractive after capex and incentive receipts are separated. We would become more cautious if support fails with estimate cuts, if inventory builds faster than sales, if the fab plan expands irrespective of orders, or if acquisition funding compromises per-share returns. The company's history and product portfolio deserve respect; they do not remove the investor's obligation to pay a sensible price. The right action today is attentive patience.

One reason to stay selective is the difference between a product moat and a stock moat. A qualified analog chip may remain in a customer design for years, but public shareholders still buy a claim on future cash flows at the market's current price. The manufacturing buildout strengthens the operating platform only if its capacity earns an adequate return; the shareholder's return also depends on entry valuation, dividends, dilution, and the time it takes for new assets to load. TI's past record supports confidence in its engineering and channel, while the present investment phase raises the hurdle for cash conversion. We

therefore treat the company as a high-quality watchlist name with a conditional action plan, not as a guaranteed compounding outcome.

The next evidence checkpoint is the next quarterly filing after the November 6 price reference. We want to see whether the Q3 strength continues across industrial, automotive, and data center; whether lead times remain healthy without inducing excess customer inventory; whether gross margin rises as loading improves; and whether capex, incentives, and operating cash flow reconcile in a way owners can understand. If these facts improve while the stock revisits support, expected return could become more attractive. If price rises faster than earnings or cash generation, patience remains the right response. That process turns the report into a living thesis with explicit evidence triggers rather than a one-time prediction.

**NARRATOR'S REFLECTION**

We are not asked to predict the next quarter perfectly. We are asked to distinguish a strong franchise from a good entry, and to change our view as cash, orders, and capital returns arrive.

**GLOBAL PARALLEL**

Across US analog specialists, European automotive semiconductor suppliers, and India's industrial electronics ecosystem, durable designs coexist with uneven customer orders. The same tension sits inside TI.

CHAPTER I

## How the Business & Stock Got Here — Past Cycles & Pivots

> *"The best way to predict the future is to create it."*
>
> — PETER DRUCKER, COMMONLY ATTRIBUTED

You and I should begin with the corporate shape, because Texas Instruments is not simply an old electronics brand attached to a semiconductor stock. It began as Geophysical Service Incorporated, a seismic exploration contractor, and became Texas Instruments in 1951. The early business married measurement, signal interpretation, and rugged electronics for difficult environments. That inheritance still makes sense when you look at the modern portfolio: TI makes components that translate physical signals into reliable electronic decisions. The company's 1950s work in silicon transistors and integrated circuits, followed by calculators and digital signal processing, repeatedly brought it to the edge of new markets. But its durable identity emerged when it stopped trying to own every end product and concentrated on selling the building blocks used by many manufacturers. The 2025 annual report now describes two reportable segments, Analog and Embedded Processing, with a smaller Other category. That simplicity is the outcome of decades of portfolio pruning, not the starting point.

The historical record also shows why a famous technical invention does not guarantee a stable business. TI made the integrated circuit commercially consequential, but it also entered consumer products, computers, calculators, defense, and communications. The home-computer episode of the early 1980s is a useful caution: consumer hardware can put a component company into price competition, inventory risk, and product-cycle exposure that it cannot control. TI exited businesses when the economics no longer matched its strengths. In the 1990s and 2000s it moved further toward semiconductors, including the acquisition of Burr-Brown in 2000, which strengthened its analog position. The more recent purchase of National Semiconductor in 2011 expanded the catalog and manufacturing base. These moves share a logic: add products with long lives, broad reuse, and a fit with TI's process technologies, then let customer designs compound over years. The portfolio is now defined more by the repeatable function of the chip than by the end device carrying the TI logo.

The stock's cycles have followed both the semiconductor cycle and the company's own capital-allocation cycle. Revenue grew from $13.37 billion in 2016 to $20.03 billion in 2022, then fell to $17.52 billion in 2023 and $15.64 billion in 2024 before recovering to $17.68 billion in 2025. The dip was not a collapse of the analog franchise; it was a correction after customers and distributors had accumulated inventory through the pandemic boom. That distinction matters to an investor because catalog chips can be ordered in advance and shipped later, creating demand whiplash. The rebound now visible in the third quarter of 2026 is sharp: quarterly sales of $6.06 billion rose 28% year over year and 11% sequentially. Such a rise is encouraging, yet one quarter cannot tell us whether a new trend rate has settled. We have to compare factory loadings, channel inventory, pricing, margins, and cash conversion over several quarters.

The company has changed the way it manufactures as well as what it sells. It historically operated mature-node fabs on 150mm and 200mm wafers; its first 300mm analog fab in Richardson began production in 2009, with a connected second fab starting in 2022. The economics are unusually relevant for analog: many useful products do not require leading-edge transistor density, so specialized process recipes and larger wafers can lower cost without chasing the newest logic node. TI has continued building 300mm capacity in Texas and Utah, while planning new shells and modular additions so equipment can be installed as demand supports it. Management's capital plan therefore creates a time mismatch: spending happens well before a new fab reaches efficient loading, while depreciation begins as production assets enter service. In a downturn, that fixed-cost base depresses gross margin; in recovery, higher utilization can create powerful operating leverage. The $4.8 billion of capital expenditures in 2025 shows that the investment phase has been real, not merely an aspiration.

The past decade's strategic emphasis has shifted toward industrial and automotive customers because their systems use many chips per product, require long qualification cycles, and often need components to remain available for a decade or more. TI's 2025 filing says roughly 75% of revenue came from industrial, automotive, and data-center markets, while its product lifecycle guidance describes typical lifetimes of 10 to 15 years or longer. Management sees that stability as a source of durable design wins. The same concentration creates a cycle risk: when factory automation, vehicle production, or capital spending pauses,

customers can draw down stock and delay orders across a broad catalog. These markets are diverse at the chip level but correlated at the macro level. The company is not insulated from recessions merely because no single customer dominates its revenue.

A parallel worth keeping in mind is Analog Devices' evolution from a specialist component supplier into a broad precision signal-chain company. ADI's 2021 acquisition of Maxim Integrated enlarged its catalog and customer reach, while TI's own organic breadth and internal fabs create a different route to scale. European suppliers such as Infineon and STMicroelectronics demonstrate another pattern: automotive and industrial qualification can support long product lives, but demand remains exposed to production cycles and policy-driven capacity investment. In India, firms such as Bharat Electronics illustrate the customer side of the same long-life procurement logic, though with a much higher defense-program mix than TI. These are not direct valuation comparables; they are reminders that a long qualification horizon protects a design after selection but does not guarantee smooth shipments. The comparative lesson is to distinguish design durability from order timing.

The balance sheet and corporate structure also show a company that has chosen to finance capacity and shareholder returns together. At September 30, 2026, the SEC filing reported $13.56 billion of debt net of issuance costs and $4.21 billion of cash plus $3.93 billion of short-term investments. Debt is material, but it is paired with a large base of recurring operating cash flow and assets that can serve multiple product generations. The firm has raised its dividend for 23 consecutive years, increasing the quarterly payout to $1.52 in 2026. The discipline question is whether returns to owners are being funded by normalized free cash flow or by temporarily strong operating conditions and public incentives during a heavy build cycle. The latest trailing-12-month report lists $9.90 billion of operating cash flow, $2.58 billion of capital expenditures, and $1.10 billion of CHIPS incentives in its adjusted free-cash-flow bridge. We should examine both gross spending and subsidy-adjusted cash generation rather than celebrating only the headline measure.

So the story that brought us here is one of deliberate specialization: broad analog and embedded products, internal manufacturing, and a direct customer interface. The historical record supports resilience, but not immunity. Revenue has swung around its trend, margins have moved with factory utilization, and strategic fabs increase the consequences of a

mistaken demand forecast. The current cycle is a fresh test of whether TI can convert product breadth and 300mm economics into per-share cash growth after an unusually capital-intensive period. When you read the remaining chapters, keep this chronology in mind. The business is strongest when end-market design wins persist, manufacturing assets are loaded, and the catalog earns pricing for long-life performance. It is weakest when customers destock while capacity costs remain fixed. That is why the price discussion must include both the recovered earnings and the capital still being absorbed into the factories.

Corporate memory is valuable in semiconductors because product decisions compound. Process nodes, package choices, test flows, and customer qualification are developed over years. A fab is not interchangeable with another fab in the way a generic office building might be; products need a stable process and can require requalification if moved. TI's long trajectory therefore created a stock of practical know-how and customer records that competitors cannot reproduce instantly. It also created organizational inertia: installed assets and legacy commitments can influence future decisions. The investor must give credit for accumulated capabilities while asking whether management is willing to exit low-return work and avoid building ahead of evidence. The company has done this repeatedly in its portfolio history, but the current capital program is its largest contemporary test of that discipline.

The old stock cycles reveal different kinds of risk. The 2008–09 recession cut demand broadly; the 2011–12 period brought industrial and electronics weakness; the 2018–19 slowdown weighed on semiconductors; COVID first disrupted supply and later triggered inventory build; and the 2023–24 period involved a more specific customer and distributor digestion. The causes differ, but the financial mechanism is similar: customer orders fall faster than a fab's fixed costs. A long product life lowers the risk of immediate obsolescence but does not smooth the number of units shipped in a given quarter. This distinction is important when assigning a premium multiple. TI may deserve a better terminal margin than a short-cycle consumer chip vendor, but its near-term earnings still follow the order cycle.

**NARRATOR'S REFLECTION**

Step back from the ticker. How did a company born in geophysical instruments become a long-lived supplier of the chips inside other people's products? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER II

## Inside Every Segment — Products, Customers & Real-World Examples

> *"The purpose of computing is insight, not numbers."*
>
> — RICHARD HAMMING

When you picture a semiconductor, you may imagine a powerful processor doing visible computation. Much of TI's business is quieter. Its analog chips connect a real-world quantity—voltage, current, pressure, sound, temperature, light, or motion—to a digital system. An amplifier raises a weak signal; a data converter turns it into bits; a power-management device routes, regulates, or protects energy; an interface chip moves signals between components. Embedded processors and microcontrollers then execute bounded control or signal-processing tasks. TI's catalog spans tens of thousands of products, many bought in low unit volumes by engineers building specialized equipment. A design might use several TI parts, and a manufacturer may buy from TI for years after the product launches. This portfolio logic supports a long tail of demand, but creates a complex quoting, stocking, and obsolescence task.

Analog is the economic center. In 2025 it generated $14.01 billion of revenue, about 79% of company sales, and $5.41 billion of segment operating profit, a 38.6% margin. The segment is split between Power and Signal Chain. Power products control energy across battery, vehicle, industrial, and data-center systems. Signal Chain products sense, condition, convert, isolate, and transmit data. A useful end-to-end example is an industrial vibration monitor installed on a pump. The equipment maker chooses a TI accelerometer or signal-chain front end to turn vibration into a clean electrical signal, uses an amplifier and converter to digitize it, and routes it to a microcontroller or gateway. Engineers test accuracy, noise, temperature behavior, and reliability, then freeze the bill of materials for production. The manufacturer pays a small per-chip price to TI directly or through a distributor, while the system vendor sells a much higher-value monitor. The chip's value is the avoided downtime and the confidence that measurements remain stable.

Power management is similarly tangible in an electric vehicle. A tier-one supplier designs a battery-management or isolated power subsystem that monitors cell voltage and temperature, balances energy, and provides safe power rails for control electronics. TI's qualified analog parts are evaluated against automotive temperature, lifetime, and functional-safety requirements. The supplier may prototype through small distributor orders, move into customer qualification, and then source larger production quantities under forecasts and supply agreements. This example explains why a low-dollar integrated circuit can remain in a vehicle platform for many years: redesign adds engineering validation and field-risk costs that can exceed any single chip price saving. It also exposes the risk: a delayed vehicle launch or a canceled platform moves volumes across a long supplier chain, and a chip maker may not see final consumer demand directly.

Embedded Processing generated $2.70 billion of 2025 revenue, roughly 15% of sales, with segment operating profit of $304 million and margin of 11.3%. The lower margin compared with Analog deserves attention. Embedded includes microcontrollers, processors, connectivity, and application-specific products; it is not simply a smaller clone of the analog portfolio. A practical example is a factory motor-drive controller. The equipment maker selects a TI microcontroller or digital signal controller, uses TI's development tools and libraries to implement motor control, qualifies the firmware against its drive hardware, and then reuses that software across product models. That reuse can increase switching costs over time. The first unit sale is only part of the economic relationship; the customer's firmware investment creates an installed base that can anchor future designs. The risk is that the customer's own software and board architecture can also make a competitor's processor more attractive if the next generation demands a different performance, cost, or connectivity mix.

A second embedded example is a connected sensor node for building automation. The OEM needs low-power control, secure wireless connectivity, and predictable availability. It evaluates an MCU with a radio, board support software, development environment, and reference design. TI's online tools, sample ordering, and distributor inventory can shorten the prototype loop; design-in engineers then support production qualification and lifecycle planning. Revenue may be recognized through direct sales or channel shipment depending on the account and geography, and prices vary with package, volume, feature set, and contractual terms. The investor should not confuse the scale of TI's product catalog with

equal profitability across SKUs. Product breadth earns the chance to be designed in; manufacturing cost, wafer size, packaging, channel mix, and support expense determine what the company keeps.

Other generated $979 million, about 6% of 2025 revenue, and $307 million of segment operating profit, including restructuring charges and other items. It includes DLP products, calculators, and custom ASIC products. A DLP projector system offers a concrete illustration: an equipment maker integrates a TI digital micromirror device with light sources, optics, image processing, and thermal controls, then sells the complete display product into education, business, or home applications. TI's contribution is a specialized optical semiconductor with a supporting ecosystem. A calculator is a simpler consumer-facing product, but it is not the company's core growth engine. Custom ASIC work can bring a customer-specific chip program, but bespoke engineering and program timing make it economically different from a catalog analog part. In 2025 the Other segment's operating margin fell sharply from the prior year, partly reflecting impairment and restructuring; that is a reason to avoid assuming all residual activities are stable annuities.

Revenue by end market is not the same as segment reporting. TI says approximately 75% of 2025 revenue came from industrial, automotive, and data center, with the remainder spread across personal electronics, communications equipment, and other markets. The company does not provide a full public segment-by-end-market profit table in the annual filing, so we should not invent one. What we can say is that industrial equipment often uses a wider mix of analog signal, isolation, power, motor control, and embedded products; automotive platforms similarly combine power management, sensing, interface, and control; data centers increase power-conversion needs and monitoring complexity. This creates multiple opportunities per design, but it also ties the overall portfolio to broad capital-investment conditions. The 2026 quarterly rebound was described by management as growth across industrial, automotive, and data center, rather than a single customer win.

Across the product families, TI sells value in four forms: electrical performance, dependable supply, lifecycle support, and engineering time saved. Parts are selected from datasheets and reference designs, tested on customer boards, and qualified to the end market's required reliability standard. Sales engineers and applications engineers assist large accounts; web

tools and distributors serve the much larger long tail. For many catalog components, a customer can compare second sources, which limits pricing power. But replacing a qualified part can trigger design changes, regulatory reviews, reliability testing, and firmware updates. The switching cost depends on the application, not just the product name. This is why you should ask whether the part is a commodity substitute or a proven function embedded in a long-lived platform. TI's breadth makes it more likely to be considered; it does not guarantee that every considered part wins.

The comparison with Europe's Infineon and STMicroelectronics clarifies the mix: all serve automotive and industrial systems, but their portfolios and power-semiconductor exposure differ. Analog Devices provides a closer comparison for precision signal chain and instrumentation, while NXP overlaps in automotive and embedded processing. In India, the growing electronics manufacturing base creates more local demand for power, sensing, and control components, but TI's economics are still primarily global and denominated in large industrial and automotive platforms. These comparisons help us interpret design content and cycle exposure without pretending that a TI analog amplifier, an Infineon power device, or an NXP automotive processor has identical economics. The company's reportable segment margins and long-life products are the best starting point for valuation, with competitors providing context rather than an automatic multiple anchor.

A final product example links the catalog to the data-center opportunity. A server or accelerator rack uses many power stages, controllers, monitors, and signal-conditioning devices to move energy safely from the facility feed to board-level loads. TI can sell power-management and sensing components into a system assembled by an OEM or contract manufacturer. The design team prioritizes efficiency, thermal behavior, reliability, and availability, then qualifies components in a system that may be revised as compute loads change. TI has described data center as a strategic end market and management included it among the drivers of Q3 2026 growth. The market may expand quickly, but power components are only one portion of a data center's total budget, and the customer mix can be concentrated among large infrastructure builders. Investors should track the dollar contribution and sustainability rather than infer a whole-company growth rate from industry headlines.

Catalog scale also changes the operating model. A TI sales team can approach the same customer through many design teams and product functions, while the customer can standardize procurement and quality systems across programs. Online product pages, datasheets, evaluation modules, and order status tools reduce the cost of discovery. This supports the long tail that a narrow high-performance vendor might not serve profitably. But small transactions can be expensive to fulfill unless automated; distribution and e-commerce help absorb that cost. Product lifecycle policy helps customers plan a decade-long equipment life, but TI must decide when to redesign, transfer, or discontinue an old product. The company's advantage therefore depends on a disciplined catalog architecture and operational execution, not simply on the number of part numbers listed.

A named, documented power example makes the value chain less abstract. Panasonic Automotive Systems, a Tier 1 supplier to automakers, used TI's LMG708B0 integrated buck converter with GaN FETs to adapt an existing premium audio design for a 48V vehicle power net. The component addresses a real constraint: higher voltage can reduce harness weight and support new electrical loads, while 48V-compatible components are less mature than 12V parts. Panasonic retained its PCB footprint and replaced the front-end power stage, reducing redesign work and time to market. TI supplies the semiconductor; Panasonic integrates it into the audio system and sells that system to automakers. The chip is sold as a component, with prototype quantities available through TI's web channel and production pricing negotiated by quantity, channel, and program. The public customer story does not disclose an automotive contract price, so we do not invent one. This is a good illustration of TI's value being measured in system redesign avoided, not just die cost.

The embedded segment has a similarly concrete but different proof point. TI's TIDM-02006 reference design demonstrates a distributed multi-axis servo drive: TMS320F280049 or F280025 real-time controllers run current loops at individual axes, while an F28388D coordinates position and speed loops and EtherCAT communication. The buyer is an industrial automation OEM designing a robot or factory motion platform; TI's reference design and MotorControl SDK reduce the engineering effort to test a control architecture. The OEM still builds its own board, power stage, safety system, and software, then qualifies the final machine. TI earns revenue when the controller is purchased for production, usually at a negotiated per-unit price that varies with volume and device grade. The public design

names the parts and engineering flow but does not identify a specific machine maker or disclose a production ASP. That distinction separates a verifiable application from a fabricated customer claim.

The Other segment's DLP activity can be followed through a named partner. In-Vision, an approved TI DLP design-house partner, develops optics and electronics and manufactures light engines for OEMs. Its PHOENIX industrial 4K UV projector uses TI's DLP670S digital micromirror device with In-Vision's pixel-shifting and optical design; the end buyer is an industrial imaging, inspection, lithography, or additive-manufacturing equipment maker that needs a projected light pattern. TI sells the DMD and related controller components into the ecosystem, while In-Vision sells the engineered optical module or complete projector subsystem. Component prices and module system prices differ; volume, resolution, wavelength, optical configuration, and support all affect the commercial quote. Here again, the ecosystem rather than a single chip creates the customer solution.

**NARRATOR'S REFLECTION**

Step back from the ticker. What is the actual product, who pays for it, and what share of the economics does it carry? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER III

## How It Is Sold & Delivered — Distribution & Unit Economics

> *"A business that makes nothing but money is a poor business."*
>
> — HENRY FORD

TI's route to market begins well before a purchase order. The company's sales and applications teams work with equipment makers, automotive suppliers, and contract manufacturers while the customer is still drawing a circuit board. Engineers search the catalog, compare electrical characteristics, request samples, test reference designs, and ask for support around power, noise, thermal limits, and software. Large accounts may be served directly; more than 80% of TI revenue was direct in 2025, according to the annual report. Direct contact helps the company learn which designs are being developed and gives it a chance to place multiple components into one platform. Distributors extend access to thousands of smaller buyers, hold inventory close to customers, and bundle logistics and credit services. TI also sells through e-commerce, where a prototype buyer can check stock and purchase a few pieces without a negotiated enterprise relationship. These routes coexist; a single design may begin through a distributor and later transition to a direct production relationship.

Once a design is selected, the customer enters qualification. Automotive programs can require extended validation across temperature, vibration, electrical stress, process change, and functional safety. Industrial customers also care about field reliability and predictable availability because replacing a failed controller can require a technician visit, a line shutdown, or a redesign of a certified system. TI's published product-lifecycle guidance says products typically remain in lifecycle for 10 to 15 years and often longer. That promise is valuable only if manufacturing, packaging, and inventory planning support it. The company uses a multi-site manufacturing footprint with internal wafer fabs, assembly and test operations, and external partners. A wafer moves through hundreds of process steps, then is diced, packaged, electrically tested, and delivered to a distribution center or customer. At each handoff, yield, cycle time, test coverage, freight, and geopolitical continuity matter. The final chip may be tiny, but the supply network is capital-heavy and global.

TI's long-term investment in 300mm manufacturing is meant to improve the cost curve and supply control. A 300mm wafer has about 2.3 times the surface area of a 200mm wafer and can produce more than twice as many chips, subject to die size and yield. TI presentations illustrate a potential roughly 40% lower wafer cost per unpackaged chip versus 200mm for a representative case; this is a company example, not a universal realized margin uplift. Internal production can protect supply and help move products between fabs, yet it concentrates fixed costs on TI's own balance sheet. A fab shell can exist before it is equipped; tools, yield learning, customer qualification, and factory loading take time. If orders lag, depreciation and labor do not disappear. If orders rise across many customers, existing capacity can be loaded and margins expand. We therefore read utilization, inventory, and capital spending together.

Here is a worked $100 revenue-to-free-cash-flow waterfall using Texas Instruments' trailing twelve months through September 30, 2026, rather than an invented product margin. TI reported $20.766 billion in revenue and $9.896 billion of operating cash flow. A deterministic calculation against those reported amounts allocates $47.65 of each $100 of revenue to operating cash flow. The company spent $2.578 billion on capital expenditures, equivalent to $12.41 per $100 of reported revenue, leaving $35.24 before incentives. TI received $1.104 billion of CHIPS Act incentives, or $5.32 per $100, resulting in $40.56 under its company-defined free-cash-flow measure; TI separately reported that measure at 40.6% of revenue. The steps show how working capital, taxes, depreciation, and factory spending sit between sales and owner cash. This is an aggregate company waterfall, not the margin of an individual component.

A better scale reference is the latest trailing twelve months through September 2026: TI reported $9.90 billion of operating cash flow and $2.58 billion of capital expenditures. Its adjusted free-cash-flow measure adds $1.10 billion of CHIPS Act incentives, producing $8.42 billion, compared with $2.42 billion a year earlier. The definition matters: GAAP free cash flow is generally operating cash flow less capex; TI's cited non-GAAP measure also adds incentive proceeds. A shareholder assessing sustainable owner cash should separate the underlying cash business from grants and tax credits, while also recognizing that a subsidy lowers the net cost of domestic capacity. The company spent heavily during the fab buildout, and capital expenditures are expected to remain elevated as it completes and equips facilities.

The cash-flow improvement is genuine, but part of it reflects the combination of a cyclical rebound, lower spending versus the prior-year peak, and incentives.

Working capital turns the revenue story into a cash story. TI holds inventory so it can deliver catalog products quickly and maintain customer service across many part numbers. The inventory strategy deliberately builds ahead for products with broad use and low obsolescence risk, but that policy consumes cash and can leave stock exposed if customers pause orders. In the 2023–24 downturn, distributor and customer inventory digestion contributed to weak shipments after earlier stocking. Receivables also move with the mix of direct accounts, distributors, payment terms, and geography. A distributor can buy inventory before end demand appears; that supports TI revenue today but pushes the eventual demand signal one step away. For that reason, book-to-bill, backlog, days of inventory, and distributor inventory commentary are leading indicators, not bookkeeping curiosities. Cash conversion can look excellent at the top of a cycle and deteriorate when products ship into channels that later need to destock.

The channel itself creates both reach and risk. Direct selling brings account insight and potentially better economics, but requires a large sales and technical-support organization. Distribution makes a broad catalog available to thousands of engineers, provides local stocking and credit, and absorbs some fragmentation in customer orders. It also means TI must manage pricing, allocations, returns, and inventory visibility across multiple intermediaries. E-commerce makes evaluation friction lower; it does not eliminate enterprise qualification. Product pricing is typically per part and varies by package, temperature grade, feature set, quantity, contract, and channel. Standard components face cross-reference competition, while specialized components may retain pricing because qualification work and failure costs matter. TI emphasizes a web catalog and direct fulfillment alongside distributors, giving it a route to monetize both prototypes and mature production demand. The mix of routes is therefore an operating choice that affects customer insight, working capital, gross margin, and the speed at which an order changes.

A useful parallel is the distribution economics of industrial automation suppliers in Europe and India. A Siemens or Schneider Electric system integrator may source sensors, power products, and controllers through a mix of direct specifications and local channel stock. An

Indian factory may depend on a distributor to get a replacement component quickly, while an automotive OEM requires a formally qualified supply chain with traceability and change control. TI sits upstream of those systems, earning a component margin rather than a complete-system margin. Its reach is a moat when an engineer can quickly find a supported part and buy globally; it becomes an exposed link if the company's availability or price loses to a qualified alternative. The margin is not simply the result of wafer cost. It is the residual after design support, inventory, logistics, manufacturing investment, yield risk, and the cost of keeping a long tail available.

For the next cycle, watch whether the business can preserve this route-to-cash advantage while the new capacity ramps. Strong indicators would include stable direct-sales share, resilient gross margin as utilization improves, inventory growing more slowly than sales, a shrinking gap between earnings and operating cash flow, and capital spending scaled to demand. A warning combination would be shipments rising while receivables and channel inventory accelerate, margins failing to expand despite better sales, or fab investment continuing without customer qualification progress. We do not have enough public detail to calculate precise customer-level DSO or product-level gross margin, so the worked waterfall remains a transparent illustration and not a measured unit economics claim. This boundary is useful: it keeps us from turning a plausible fab advantage into unsupported precision.

The waterfall definition matters. Conventional operating cash flow less capital expenditures was $7.318 billion before CHIPS receipts; TI's non-GAAP figure adds $1.104 billion of incentives and reports $8.422 billion of trailing free cash flow. That incentive changes the conversion rate by a visible amount and should be shown, not hidden. The ratio to revenue uses the company's published trailing revenue of $20.766 billion; all displayed per-$100 values were calculated mechanically from those reported figures. The ratio is not a forecast, a product-level contribution margin, or a guarantee that an unusually high recovery-year cash conversion will persist. A shareholder should normalize it across inventory cycles and the fab buildout.

Supply continuity is itself part of the product. Industrial and automotive customers want to avoid a single component shortage shutting down a production line. TI's internal capacity and multisite manufacturing can lower dependency on external foundries, while

geographically distributed assembly and test operations provide additional flexibility. This advantage can attract a customer before the part is designed in, because the procurement team is evaluating supply assurance as well as price. But the same network has concentration risk in specialized equipment, materials, utilities, and skilled labor. The global footprint does not remove geopolitical risk; it gives management more options to route production if a disruption is contained. The eventual proof is delivery performance through a real shortage or demand spike, plus steady economics across cycle phases.

**NARRATOR'S REFLECTION**

Step back from the ticker. How does a chip move from a fab and design win into collected cash, and what can interrupt that path? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER IV

## The Battlefield — Rivals, Switching Costs & Moat Reality

> *"In the long run, the only sustainable competitive advantage is your ability to learn faster than the competition."*
>
> — ARIE DE GEUS

TI's moat is not one patent or one flagship chip. It is the combination of a broad catalog, process technology, manufacturing capacity, global channels, and many long-lived customer designs. That combination can reduce cost and improve availability while giving engineers a high probability that a part will remain orderable. The company says it has tens of thousands of products and a large installed base across diverse applications. The breadth creates repeated opportunities to place several parts in a design, and it gives TI data about customer needs across end markets. But breadth also carries a catalog-management burden: hundreds of thousands of customer programs can depend on old products, and those products must be supported without wasting capital on every possible SKU. Scale is most defensible in general-purpose analog, where design support, a broad parametric portfolio, and affordable manufacture matter as much as peak performance.

Manufacturing is the most distinctive part of TI's claim. The company has chosen to own most of its wafer manufacturing and assembly/test rather than rely mainly on external foundries. In its 2025 filing, TI described a goal for more than 90% of wafers, assembly, and test to be manufactured internally. This can improve supply assurance and process control, and 300mm wafers can lower cost for suitable products. The tradeoff is capital intensity. Competitors with fab-light models can redirect cash to R&D, buy capacity only as needed, or avoid carrying underutilized plants. TI's model can win when its fabs are loaded and its process portfolio fits mature-node analog and embedded products; it can lose on return on capital when capacity outpaces customer demand. The advantage is therefore conditional on utilization, product migration, yield, and capex discipline. We should not treat ownership itself as proof of lower total cost.

The second durable edge is customer design persistence. A microcontroller or analog front end that works in a safety-related or long-lived industrial product can remain qualified through multiple product versions. Customers are reluctant to change a tiny component if the change forces new software validation, board design, manufacturing tests, and regulatory documentation. The switching cost is highest where the component is embedded in a complex control loop or must be supported for years. It is lower for a standard power regulator with multiple pin-compatible alternatives. TI's lifecycle statement of typical 10- to 15-year availability supports this positioning, but the commitment is only as strong as the company's ability to maintain process and packaging flows. Long product lives also make old nodes and older tools economically useful, which differs from leading-edge compute where rapid node transitions dominate.

The third edge is the market channel. TI's direct relationships with larger customers provide visibility into programs and design activity, while distributors extend its reach to fragmented engineering teams. Online ordering and product documentation make it easier for a new customer to test a part without a lengthy sales process. These routes create a feedback loop: a designer finds a product, uses reference software or a development board, orders samples, then moves the part into production. That is a real commercial advantage when TI's support is responsive and the part is available. However, competitors have capable direct teams and broad distributor networks too. A part can be designed out when a rival offers better performance, a lower total system cost, a more favorable package, or a second-source guarantee that eases supply risk. The channel brings TI into the consideration set; engineering fit closes the sale.

Rivals include Analog Devices in precision signal chain, NXP and STMicroelectronics in many automotive and embedded applications, Infineon in power and automotive control, Microchip in microcontrollers and industrial embedded markets, and numerous specialist analog firms. Competition is product-by-product. TI can have a strong general-purpose portfolio and still lose a high-performance converter socket to a more specialized vendor. Conversely, TI can win with a catalog device whose performance is adequate and cost is attractive. Market share claims need careful scope: semiconductor revenue rankings combine memory, logic, analog, and discrete devices, which are not useful measures for a specific function. The annual report does not disclose a full market-share table for each product

category, so we rely on the customer and product evidence rather than borrowing an industry-wide ranking as proof.

A European comparison helps separate the durability of industrial applications from the economics of a supplier. Infineon and STMicroelectronics both sell into automotive and factory systems with qualification cycles, but their exposed product mixes include power devices and automotive controllers that differ from TI's analog-heavy catalog. Analog Devices and Maxim's combined portfolio has meaningful signal-chain breadth and a strong applications-engineering presence. NXP competes in vehicle networks, processors, and embedded platforms. In India, a public-sector procurement chain such as BEL's radar and communications programs values reliability and long qualification, while the channel and end-market concentration differ greatly. Those parallels support the idea that customer qualification can be sticky; they do not establish that TI is insulated from pricing competition or the global demand cycle.

The strongest moat test is not whether TI speaks about a structural cost advantage, but whether customers and financial results confirm it. In 2025 analog margin recovered to 38.6% from 37.9%, while embedded operating margin slipped to 11.3% from 13.9%. Total gross margin declined to 57.0% from 58.1%, as the company cited higher costs from capacity expansion, partly offset by better factory loadings. Those outcomes say the asset base is still in transition. In Q3 2026, sales rose sharply and operating profit grew faster than revenue, a positive sign of leverage. The next test is whether that leverage persists after demand normalizes and the new fabs add depreciation. A durable moat should show up in sustained profitability across cycle phases, not only in a good quarter.

The main vulnerabilities are capital-cycle timing, standard-product competition, end-market concentration, and execution in the capacity roadmap. More internal manufacturing can improve supply independence but also increases exposure to labor, tool availability, utilities, and regional disruptions. A broad catalog can mean better cross-selling but also more inventory and lifecycle obligations. Customer qualification makes current designs sticky but does not prevent a customer from shifting a future platform. Data-center power demand is a compelling opportunity but can be lumpy and influenced by a small number of large infrastructure builders. Automotives and industrial customers are diversified by product but

may respond together to macro downturns. The moat is real, but it is not a force field; it converts engineering history and manufacturing scale into a better starting position, with margins and returns determined by how wisely capacity is deployed.

For your own follow-up, the External Learning Hub should include the 2025 10-K for product mix and risk factors; the 2026 Q3 release and prepared remarks for current demand and capital spending; TI's capital-management deck for management's 300mm cost case; the worldwide manufacturing and Richardson pages for factory geography; product category pages and lifecycle policy for the customer interface; the SEC 10-Q for debt, acquisition, cash, and inventory; and peer annual reports from ADI, NXP, ST, and Infineon for the rival context. Read management's fab claims as hypotheses and look for realized margins, utilization, and per-share cash growth in later filings. That is a more disciplined way to learn the business than memorizing a slogan about scale.

Switching cost needs a customer-level lens. If an engineer uses a standard low-voltage regulator in a new consumer accessory, the competing part may be easy to substitute before board layout is fixed. If that regulator is in a certified medical monitor or a vehicle subsystem with a long production program, the replacement may require thermal testing, safety review, sourcing approval, software changes, and renewed lifecycle commitments. TI's product longevity can lower uncertainty for the second customer, but only if the exact part remains available and meets the next revision's requirements. A supplier can have a broad moat at portfolio level and weak pricing power on individual SKUs. This is why we value application relationships and production persistence, while treating broad catalog size as a qualified advantage rather than proof of monopoly economics.

An additional competitive risk is that the industry's economics are shifting around data-center capital expenditure. High-performance compute receives the headlines, but foundational analog and power parts may benefit from increasing system complexity and load. The opportunity is attractive because TI's process technologies and power portfolio are relevant; the uncertainty is whether system architectures standardize around a small number of suppliers or disperse parts among multiple qualified vendors. Large customers can negotiate aggressively and may second-source critical components. TI's broad reach creates many individual sockets, but each one can still be contested. The company must keep

investing in product development and applications support while ensuring that the fab footprint does not force it to chase volume at uneconomic prices.

**NARRATOR'S REFLECTION**

Step back from the ticker. Which parts of TI's advantage are structural, which are cyclical, and where can rivals take the design? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER V

## Management on Trial — Leading KPIs & Walking the Talk

> *"You cannot manage what you cannot measure."*
>
> — PETER DRUCKER, COMMONLY ATTRIBUTED

The first leading indicator is customer demand before it becomes revenue: orders, backlog, book-to-bill, and lead times. TI does not disclose a complete public backlog by product family, so the third-quarter call's comments about broad growth and modestly lengthening lead times are directional rather than a substitute for an order table. In an analog business with a long catalog, customers can place replenishment orders in waves. Bookings may recover before shipments, and distributor purchases may not equal final usage. We therefore look for confirmation in revenue from industrial, automotive, and data center together, rather than treating one fast-growing pocket as the whole story. When management says demand is broadening, the check is subsequent quarterly sales, pricing, and customer inventory commentary. A sharp growth rate is good evidence that the cycle turned; it is not yet proof of a durable mid-cycle growth rate.

The second KPI is the end-market mix, especially industrial, automotive, and data-center exposure. TI says these markets represented roughly three-quarters of revenue in 2025, and the company has made them a strategic priority because semiconductor content per system can expand over time. Industrial equipment digitization and electrification increase the need for sensing, signal conditioning, motor control, power conversion, and embedded control. Cars add battery, power, sensing, connectivity, and safety electronics. Data centers require power delivery and monitoring around processors, storage, and networking. These demand drivers operate over years, but equipment orders are cyclical and platform designs can shift. We should ask not only whether the long-term silicon content rises but how much profit and cash TI earns after it builds capacity to supply that content. The 2026 recovery in industrial, automotive, and data center is promising; the next annual filing should reveal whether growth broadened further.

Third, track factory utilization and gross margin. Revenue can rise while margins fall if new factories are underloaded, if depreciation grows ahead of output, or if product mix is unfavorable. In 2025 total gross margin was 57.0%, down 1.1 percentage points from 2024. TI attributed pressure to capacity-expansion costs, partly offset by improved factory loading. Analog segment margin increased modestly to 38.6%, while Embedded Processing margin contracted to 11.3%. These are not just backward-looking scorecard figures: they tell us whether the manufacturing strategy is being absorbed by demand. With semiconductor fabrication assets, utilization interacts with product yield and fixed costs. A few quarters of stronger loading can create margin recovery, but a large new fab takes time to reach mature output. We should compare margin gains against capex, incentives, depreciation, and the required return on invested capital.

Fourth, measure inventory quality and channel health. TI intentionally holds inventory to support availability across a broad catalog and long customer programs. The key is not simply to minimize inventory; a chip that is unavailable when a factory needs it can damage a customer relationship. Instead, ask whether inventory rises in line with expected orders, whether aging or obsolescence reserves move, and whether distributors are holding stock that customers have not consumed. The 2023 and 2024 revenue contraction shows how a prior build can create a protracted digestion period. On the Q3 2026 call, the company described demand as broadening and lead times as slightly longer. That is consistent with tightening supply, but it can also encourage the next round of customer over-ordering. A healthy cycle should show customer demand gradually catching up with shipments, not a temporary rush to secure scarce capacity.

Fifth, watch capital intensity and incentive-adjusted cash generation. TI's objective is long-term free cash flow per share growth, and its capital plan has been unusually large because of domestic fab construction. The company reported $4.8 billion of capex in 2025. Over the 12 months to September 2026 capex was $2.58 billion, while the company recognized $1.10 billion of CHIPS Act incentives in its non-GAAP free-cash-flow calculation. The Q3 2026 call set a $2 billion to $3 billion capex range for the year and management leaned toward the high end. The relevant test is not whether management spends less in one year; it is whether the completed factories produce enough incremental volume and cash to justify the total

capital consumed. We should track annual capex, cash incentives, depreciation, operating cash flow, dividends, repurchases, and per-share measures.

Sixth, review design-win durability and new-product contribution. TI does not report revenue from design wins as a standalone KPI, but applications support, new part introductions, product lifecycle, and software reuse can indicate whether its portfolio remains relevant. In embedded processing, a customer may reuse firmware and development tools through several generations, increasing switching costs. In analog, a new power or sensing design may create many small sockets across a platform. The company's broad catalog lets it participate in more projects; the pace of new product additions and adoption matters more than the raw product count. A useful management question is whether R&D spending is generating products that win in faster-growing applications, or merely maintaining legacy parts. TI invested around $2.0 billion in R&D in 2025, a meaningful commitment but one whose return is observed over long customer-development periods.

The walk-the-talk audit is mixed but coherent. Management has repeatedly described 300mm as a cost and supply advantage, and the physical capacity plan is progressing: RFAB2 is in production, Sherman SM1 began production in late 2025, and other sites are being prepared. Yet the expected payoff is delayed, and 2025 gross margin still reflected expansion costs. Management's long-term ambition is free cash flow per share growth; trailing cash flow improved substantially by Q3 2026, but its definition adds CHIPS incentives and it coincided with a cyclical rebound and lower capex versus the preceding period. The claim of broad product diversity is supported by the lack of a single dominant product, but end-market exposure is concentrated in industrial, auto, and data center. The company's promises are not obviously contradicted; they remain under proof. This is a case where capital discipline should be graded over a full cycle, not an isolated quarter.

The announced Silicon Labs acquisition is the newest test of management's capital allocation. TI agreed to pay $231 cash per Silicon Labs share, with closing expected in the first half of 2027 subject to approvals; the 2026 SEC filing described a $5 billion delayed-draw facility available to support the transaction. Strategic rationale includes adding low-power wireless connectivity and embedded products, which could deepen TI's MCU and IoT portfolio. But the price is paid before integration benefits are proven, and the deal brings

financing, regulatory, integration, and potential dilution-to-cash-return tradeoffs. For a company whose stated model prizes disciplined allocation and long-run FCF per share, the bar is clear: show that acquired engineers and products expand design wins, preserve returns, and do not crowd out internal capital projects or shareholder returns. The acquisition should be judged on post-close cash generation and customer cross-sell, not the strategic story alone.

Management's credibility will rise if it reports three things consistently: factory loading and the timing of depreciation absorption; capital expenditure and CHIPS-related receipts with a reconciliation to GAAP operating cash flow; and product/customer trends that establish whether growth is broad. It will fall if capacity plans expand despite weak orders, if free cash flow is presented without separating incentives, or if acquisition economics remain opaque. We should also track board-approved dividend growth against underlying cash generation. A 23-year streak demonstrates a shareholder-return preference, but does not guarantee future yield safety if a downturn and spending peak overlap. This operating dashboard keeps the analyst's attention on facts that precede earnings and on whether management's actions match its declared aim of per-share cash growth.

Another management KPI is employee and engineering capacity. Long-term product success depends on applications engineering, process integration, quality systems, and software support, not just factory buildings. The 2025 annual report described continued investment in R&D and capacity, while the Q3 2026 transcript included commentary on tool lead times and customer demand. If external equipment delivery slows, capacity may arrive later than planned; if engineering teams are stretched across an expanding catalog and acquisition, product launches may slip. TI does not publicly report a compact KPI that links engineering headcount to successful design wins, so we have to infer progress from R&D expense, product releases, customer qualification, and the eventual revenue mix. The Silicon Labs transaction could strengthen embedded connectivity but also requires management attention during a capital-intensive period.

The right executive scorecard should include absolute dollars and per-share outcomes. Rising revenue can coexist with falling returns on capital, and rising total cash flow can coexist with flat free cash flow per share if share count, capex, or acquisition funding grows. Management's public framing around FCF per share is useful because it asks whether the

owner's claim on cash grows after investment and capital allocation. We should compare the stated ambition with reported cash flow after capex, incentive receipts, interest, and diluted shares. The measure is not a quarter-by-quarter promise; it is a through-cycle test. A business that spends heavily now may still create value if the assets earn returns for decades, but investors should demand evidence of that conversion rather than infer it from the size of the investment.

On the Q3 2026 call, CFO Julie Knecht said the quarter's margin was "driven by higher revenue," while also identifying pricing, loading, and mix as contributors. That short comment is consistent with the reported 73% year-over-year operating-profit increase on 28% revenue growth, but it does not prove that the same incremental margin will persist. Knecht also reported inventory of $4.7 billion, or 194 days, down two days sequentially. Those figures give investors a tangible cycle check: demand recovered, yet inventory remained large in absolute terms and should be watched against shipment growth. In her Q&A, management described long-run incremental gross-profit flow-through expectations after depreciation; investors should treat that as guidance to test against actual margins rather than as a guaranteed result.

**NARRATOR'S REFLECTION**

Step back from the ticker. What measurements should lead reported earnings, and what has management delivered against its stated plan? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER VI

## Following the Cash — 10-Year Financials & Earnings Quality

> *"Revenue is vanity, profit is sanity, but cash is reality."*
>
> — COMMON BUSINESS MAXIM

The ten-year revenue series is a cycle, not a straight line. TI reported approximately $13.37 billion in 2016, $14.96 billion in 2017, $15.78 billion in 2018, $14.38 billion in 2019, $14.46 billion in 2020, $18.34 billion in 2021, $20.03 billion in 2022, $17.52 billion in 2023, $15.64 billion in 2024, and $17.68 billion in 2025. Each figure comes from the annual filings; the pattern reveals a company that grew its revenue base over the decade but still experienced both macro shocks and an inventory cycle. The 2019 pause preceded COVID; 2021–22 brought unusually strong demand; 2023–24 retraced; and 2025 began recovery. Annual sales alone conceal the contribution of acquisitions, product mix, pricing, and customer inventory. The investment case therefore depends on normalized earnings and cash over a full cycle, rather than extrapolating either the 2022 peak or the 2024 trough.

The profit line moved more sharply than revenue because the factories carry fixed cost and because taxes and product mix can shift. Net income rose from about $3.60 billion in 2016 to $8.75 billion in 2022, then fell to $6.51 billion in 2023, $4.80 billion in 2024, and $5.00 billion in 2025. EPS also reflects the share-count trajectory and tax effects. The 2025 gross margin was 57.0%, while operating margin was 34.1%; in Q3 2026, operating profit grew 73% year over year on 28% sales growth, a powerful operating-leverage quarter. The fact that operating profit grew faster than sales is positive, but annual results must show whether margin recovery continues as capacity expands. A single quarter can be helped by fixed-cost absorption and mix; it cannot establish the mature return on the latest fab investments.

Segment quality differs. Analog supplied $14.01 billion of 2025 sales and $5.41 billion of operating profit, with a 38.6% segment margin. Embedded Processing supplied $2.70 billion but only $304 million of operating profit, an 11.3% margin. Other contributed $979 million and $307 million of operating profit, though that figure includes restructuring and special factors. The consolidated margin is therefore dominated by Analog. If embedded

connectivity and processors scale, they could diversify the earnings base and create software-related switching costs; if their margins remain structurally lower, revenue growth there will not translate proportionately into profit. The Silicon Labs deal makes this question more important. Investors should evaluate the acquired product portfolio against the company's ability to keep Analog's economic engine healthy, rather than treating product additions as automatically accretive.

Cash flow needs a bridge between earnings and fab investment. At September 2026, trailing operating cash flow was $9.90 billion, compared with $6.90 billion for the prior-year period. TI reported $2.58 billion of capex over that period and $8.42 billion of adjusted free cash flow after adding $1.10 billion of CHIPS incentive receipts. For the preceding 12 months, capex had been $4.82 billion and adjusted free cash flow was $2.42 billion. This is a remarkable improvement, but the free-cash-flow series is sensitive to the timing of capex, tax incentives, working capital, and the industrial recovery. A normalized owner-cash estimate should show both operating cash flow less capex and TI's incentive-adjusted measure. It should not capitalize the highest cash figure as though the factory build is over. The 2025 annual filing reported a large investment year, and management expects capital spending to remain elevated through the expansion.

Working capital is a quiet source of volatility. Inventory supports immediate availability for a broad catalog and helps customer continuity, but it ties up funds and can become stale. Accounts receivable move as shipments and payment terms shift, while payables and customer advances can partially offset the cash requirement. The business model's broad SKU count and channel network make product-level tracking difficult from public consolidated accounts. The 2026 Q3 filing shows the balance sheet and cash flows, but it does not provide enough detail for a precise customer DSO series by segment. We should not invent one. The correct forensic approach is to compare inventory and receivables with sales across annual periods, read inventory reserve disclosures, and assess whether CFO tracks net income through recovery and downturn. The 2023–24 cycle is a useful natural experiment: inventory normalization and factory loading impacted the cash and margin story even though long-term design positions remained in place.

The balance sheet is sound enough to carry investment, yet leverage is not zero. At September 30, 2026, cash and cash equivalents were $4.21 billion and short-term investments were $3.93 billion; debt including issuance costs was $13.56 billion. The company had an undrawn revolving facility and no commercial paper outstanding, while it also established up to $5 billion of delayed-draw financing for the Silicon Labs transaction. Debt maturity ladder and coupon structure are manageable in an investment-grade framework, but the acquisition adds cash and funding requirements. The balance sheet should be read alongside annual dividends, capex, and integration outlays. The business is not in a liquidity crisis, and its cash generation is strong; the question is whether incremental borrowing and acquisition cost are matched by sufficient incremental cash earnings.

Shareholder returns include dividends and repurchases. TI raised its quarterly dividend to $1.52 in 2026, an annualized $6.08 per share, and the company's investor materials emphasize long-run free cash flow per share. Buybacks can increase per-share value when shares are repurchased below intrinsic value, but they can also simply offset compensation dilution or consume funds at elevated valuation. We should check shares outstanding year over year, not assume repurchases always create value. TI's share count has generally benefited from buybacks over longer horizons, but a large acquisition financed with cash and debt shifts the capital-allocation mix. The cash dividend is a visible commitment; it is not evidence that the current share price is cheap.

ROIC and ROE need careful interpretation in a fab cycle. A high return metric in years of mature plants can fall when a company builds new capacity before sales catch up. Accounting book value may not capture the full value of process knowledge or customer design positions, and tax credits can reduce cash cost. Conversely, a strong brand cannot rescue poor incremental returns if new fabs sit underused. Finviz's contemporaneous card showed a 22.0% ROIC and 39.5% ROE, but these are vendor-computed trailing metrics rather than audited forecasts; TI's published financial statements remain the primary record. We should compare returns on incremental capital after 300mm investments become productive, and we should separate pre-incentive capital returns from subsidized cash economics when evaluating policy resilience. The main evidence still to come is whether mature capacity produces enough volume to offset depreciation and financing costs.

The cash-quality verdict is therefore favorable with a capital-cycle reservation. Earnings have been supported by real cash generation, the current recovery is visible in reported results, and the debt burden is backed by liquidity and cash flow. But recent adjusted FCF is boosted by incentive receipts, the build program has shifted the timing of capex, and the 2023–24 downturn reminds us that customer inventories can reverse. TI is not a capital-light compounder in the present phase. You and I should underwrite mid-cycle operating cash, deduct the capex needed to sustain and complete capacity, and then assess dividends, debt repayment, buybacks, and acquisition funding. Until those pieces converge over several years, the latest cash-flow surge should be treated as an encouraging data point—not as a new normal guaranteed by the business model.

The decade's peak-to-trough sequence offers a stress test for normalized cash. Revenue increased materially from 2016 to 2022, but the most recent cycle demonstrated that profitability does not rise smoothly with installed product breadth. The company's 2025 gross margin was still below prior peak-cycle levels, while the new manufacturing base added costs. The relevant question is not whether TI can return to the best historical gross margin; it is whether the business can produce attractive cash returns at more ordinary utilization. A conservative model would examine a mid-cycle sales level, margins below the peak, and capex that includes maintenance plus the remaining strategic expansion. If those assumptions still support dividend coverage, debt service, and incremental investment, the thesis is robust. If only peak utilization supports the valuation, the risk is higher.

Capital allocation has four competing calls on cash: dividends, repurchases, organic fab investment, and acquisitions. Dividends provide a visible return and a commitment to owners, but reduce the cash available for capacity. Repurchases can shrink the denominator but should be evaluated against price. Fab investment can build a cost and supply advantage, but needs years to mature. Acquisitions may speed portfolio growth, yet add integration and debt. At September 2026, the planned Silicon Labs transaction was expected to close in the first half of 2027 and would be funded with cash on hand and debt financing. This creates a period in which investors should scrutinize whether the company maintains its stated discipline or allows each strategic rationale to justify a larger balance sheet. The financial position is strong enough to act, but strength does not make every deal attractive.

**NARRATOR'S REFLECTION**

Step back from the ticker. What does a full-cycle record say about growth, margins, cash conversion, leverage, and dilution? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER VII

## Why the Stock Trades Here — Market Expectations vs. Peers

> *"Price is what you pay; value is what you get."*
>
> — WARREN BUFFETT

At the verified November 6, 2026 close, the live consumer snapshot placed TXN at $295.73, with market capitalization of about $270 billion, trailing P/E of 38.37, forward P/E of 25.47, price-to-sales of 13.01, and price-to-free-cash-flow of 36.93. The snapshot also showed a 1.95% indicated dividend yield and a 52-week range of $152.73 to $334.03. These are timely market measures rather than audited operating facts. They describe a premium valuation for a company with a quality catalog, strong recent rebound, and a multi-year manufacturing buildout. The 84% one-year price gain reported in the same snapshot means the market has already repriced a substantial share of the recovery. When you ask why TXN trades here, the answer is not only its historical margins; investors are paying for dependable long-life products, internal manufacturing, dividend growth, and an expected return to stronger cash earnings.

The 2025 annual report reveals that earnings are still below their 2022 level: net income was $5.00 billion compared with $8.75 billion in 2022, even as sales recovered to $17.68 billion. The latest 12-month snapshot through Q3 2026 shows $20.77 billion of sales and $7.07 billion of net income, but those results include a strong upturn and do not yet represent a complete fiscal year. The 38.37 trailing multiple is therefore measured on recovering earnings; a forward multiple in the mid-20s assumes earnings growth continues. If factory loading and prices improve, the trailing ratio can fall quickly without share-price decline. If the cycle stalls, the denominator may disappoint and the multiple can compress. The investor's task is to judge both the earnings trajectory and the quality of the forward estimate, not to declare a stock cheap from a lower forward P/E.

A reverse-valuation lens starts with the price and asks what must be true. At roughly $270 billion of equity value, the market expects TI to preserve high returns on a broad catalog, grow data-center and automotive content, bring new 300mm capacity to productive use, and

eventually convert expansion spending into cash per share. The price-to-sales multiple near 13 is demanding for a cyclical semiconductor manufacturer; it implies confidence in profitability and durability. The price-to-free-cash-flow ratio near 37 is also elevated, though trailing cash flow is distorted by the recent cycle and fab spending. We should not present a single-point reverse DCF with assumed discount rates and long-run growth as if it were a fact. Instead, these multiples tell us that a mediocre outcome could deliver disappointing returns even if TI remains an excellent business. The market expects sustained execution, not merely a return to peak revenue.

Peer comparison is useful but needs segment-aware context. Analog Devices is the closest broad analog peer, with signal-chain and power products and an expanded catalog after Maxim. NXP competes in automotive and embedded systems; STMicroelectronics and Infineon have broader power and automotive semiconductor exposure; Microchip is a major embedded and industrial supplier. Each has different memory exposure, wafer ownership, gross margin, growth, debt, and capital intensity. A one-line P/E ranking can therefore mislead. TI's 2025 Analog margin of 38.6% and long-life catalog support a premium over lower-return or more volatile peers, while current capex and lower Embedded Processing profitability argue for scrutiny. The appropriate question is what return on incremental capital TI can earn after the current buildout compared with what peers can earn without matching its fixed-cost burden.

Valuation history should also include the business's cyclicality. The shares have repeatedly traded at premium multiples because investors value long product lives and a disciplined dividend, but the earnings base can move substantially when industrial and auto demand changes. In 2022, annual sales and net income reached $20.03 billion and $8.75 billion; by 2024 they were $15.64 billion and $4.80 billion. The stock's one-year 84% gain into November 2026 captures a rebound from a depressed 2025 reference period and a broader semiconductor re-rating. A high-quality franchise can be overpriced; a cyclical recovery can be underpriced early and overpriced late. We do not have a full, consistently sourced 20-year monthly valuation series in this run, so we avoid stating a precise historical P/E band. The annual financial record and current snapshot support the more important point: current valuation assumes above-trough earnings.

A three-outcome framework is more honest than a point estimate. In the bull case, industrial and automotive content grows, data-center power demand adds incremental sockets, 300mm production improves the unit-cost mix, and free cash flow per share grows after capex moderates. In the base case, sales normalize to moderate growth after a cyclical rebound, margins recover but are partly offset by depreciation, and the stock's premium limits annualized upside. In the bear case, customers destock, fabs remain underloaded, the acquisition consumes financing capacity, and valuation compression compounds falling earnings. The price appears to require a high probability of the first two cases and little tolerance for the third. The provided technical snapshot's own valuation label reads "Reasonable / In-line," but that classifier is not a discounted cash-flow appraisal. Our independent judgment must weigh high quality against a premium market price and unresolved capital returns.

For peer anchoring, compare quality rather than just multiples. Analog Devices offers strong signal-chain breadth and likely shares the long qualification dynamics; NXP and ST have larger automotive connectivity and microcontroller exposures; Infineon has power-semiconductor scale; Microchip has broad embedded control. TI's internal manufacturing and 300mm program give it a differentiated fixed-cost profile. That may justify a better multiple when utilization is high and the fabs are competitive; it may justify a discount when returns on the buildout are below expectations. The cross-market analogy to European industrial suppliers is that the market often prices resilient customer positions at a premium until a cycle downturn reveals the operating leverage. The Indian analogy is the contrast between a long-order-cycle defense or industrial supplier and its listed market multiple: visible demand is valuable, but price still determines expected return.

My valuation conclusion is not that TI is uninvestable. It is that the margin of safety is thin at $295.73 given the trailing cash-flow multiple, the sharp prior-year rally, and the need to prove that new capacity yields acceptable cash returns. The useful entry trigger is a pullback into the snapshot support band, but even that would not make valuation attractive by itself. A better risk/reward would combine technical stabilization with another quarter of broad organic demand, sustained margin improvement, and transparent capex/incentive reconciliation. If price falls while fundamentals remain intact, the investor can reassess; if

price keeps running before earnings catch up, waiting is rational. The company's quality deserves attention, but the current price asks us to be patient and selective.

A quality premium is rational when it reflects repeatable cash generation, not when it is simply a reward for familiarity. Investors often trust TI because the business has been profitable across many product cycles and its dividend record is unusually long. That trust can lower the return demanded by the market and raise the multiple. However, the semiconductor sector's capital cycle can challenge even trusted firms. If a high multiple assumes continued earnings growth and the latest recovery fades, the stock can fall while the company remains financially healthy. This is the key distinction between business risk and valuation risk. Our stance is cautious on expected return, not on solvency or product relevance. We can admire the company and still ask for a better entry price.

Reverse DCF is best treated as a question set. What sales growth must persist to support the current price? How much of that growth can convert into gross profit after depreciation? What level of maintenance and expansion spending is necessary to keep the product portfolio supplied? Will the acquisition increase cash per share after financing? We do not have enough verified, consistent inputs here to publish a single implied growth rate. But the high price-to-sales and price-to-cash-flow ratios already tell us that the market is pricing a strong outcome. If the answers to these questions improve with each quarterly filing, the premium can be earned. If management's outlook depends on incentives, high utilization, and a successful acquisition all arriving at once, the margin of safety is modest.

> **NARRATOR'S REFLECTION**
>
> Step back from the ticker. How much recovery and long-run quality are already embedded in the price? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER VIII

## Reading the Tape — Trend, Volume & Momentum Opportunity

> *"The trend is your friend until the end when it bends."*
>
> — ED SEYKOTA, COMMONLY PARAPHRASED

The technical read is sourced from the fresh TXN snapshot retrieved November 9, 2026, whose reported close of $295.73 corresponds to the latest completed US session, Friday November 6. The snapshot classifies the trend as Stage 2 and the technical timing gate as CLEAR_FOR_CONSIDERATION. The close sits above the reported 50-day average at $279.80, 200-day average at $262.78, and 30-week average at $285.84. Its 10-week EMA is $284.42, and daily RSI is 52.75. These values are precomputed by the TradingView consumer snapshot and the Stage 2 classifier; no indicators are estimated here. The technical picture is constructive: price is above key trend references and is not in an extreme RSI state. But one close cannot confirm a durable trend, and the industry dashboard reported only 26% of semiconductor peers above their 200-day EMA. Individual strength is ahead of weak sector breadth.

The snapshot reports a 1-year gain of 84.39% and a position 11.47% below the 52-week high of $334.03. On the day, the quote rose 6.62%, with volume about 1.33 times its 30-day average according to the data card. That burst can mark renewed sponsorship, but it followed a volatile period and is not sufficient evidence of institutional accumulation by itself. The 1-week performance was negative in the snapshot, while one-month and three-month returns were modestly positive. This mixed short-horizon tape suggests recovery with chop rather than a clean breakout. We should watch whether the price holds above the $292.56 nearest support listed by the classifier, whether it can retake and hold the $300–$319 area, and whether up weeks continue to carry stronger volume than pullbacks. The TradingView rating was mildly positive, but an indicator score does not replace price structure.

The snapshot's Stage 2 evaluation gives an entry posture of BUY_AT_CMP_NO_FOMO, support band of $280.15 to $296.95, and invalidation stop of $273.39. It also records extension of about 4% above the 10-week EMA and 5.69% above the 50-day average, below

the stated 10% ceiling for immediate consideration. Our final recommendation is more cautious than the mechanical timing label because the business and valuation review does not establish enough margin of safety at a premium multiple. The technical band is therefore a monitoring zone, not an order. A close below $273.39 would invalidate this snapshot's tactical setup; a sustained hold near the rising averages and improving sector breadth would restore the short-term case. Since earnings are quarterly and price moves daily, these reference levels require a fresh snapshot before any future action.

For a momentum trader, the setup has three positives: a clear Stage 2 designation, price above the 200-day and 30-week references, and a recent high-volume advance. The negatives are equally concrete: the stock remains below its 52-week high, the semiconductor industry breadth is weak, and the last year's gain is already large. No breakout type was recorded. The latest advance may be a recovery leg after a downcycle rather than a new long-term leadership phase. To earn an aggressive entry, the stock would need to consolidate above support, build relative strength against the S&P 500 and semiconductor group, and break resistance with follow-through. To earn an investment entry, it would also need earnings and cash-flow evidence. Technical and fundamental gates are separate: one can improve while the other deteriorates.

The short-term trade horizon is one to twelve weeks. A trader should define the risk before entry, use the snapshot's support and stop only as provisional references, and avoid chasing a gap after an earnings-related move. The base case is range behavior while the market digests the sharp gain; a bullish continuation requires a sustained close above the recent resistance zone with sector participation; a bearish outcome is a failed support retest and a break below the invalidation level. The stock's dividend yield does not protect a short-term position from a 10% or larger drawdown. A position sized for a business-quality thesis may be too large for a tactical trade, so the time horizon and exit rule need to be explicit.

The medium-term horizon is three to eighteen months. The cycle can keep improving if industrial orders, auto demand, and data-center power content all remain positive. But the sector's low breadth reading warns that the recovery is not broad across listed semiconductor names. The 2026 results may pull in earnings estimates, while the acquisition and fab spending influence expectations beyond the immediate quarter. Medium-term investors

should follow relative strength, the 30-week trend, next-quarter guidance, and the next 10-Q's cash and inventory disclosures. If the stock remains above the rising 30-week average while earnings rise, the trend thesis strengthens. If the price breaks that level as estimates fall, the market may be signaling that the recent earnings beat was cyclical rather than structural.

Long-term investors should treat the chart as an entry discipline, not a business-quality model. A strong long-term compounder can experience multiyear periods of underperformance when bought at an elevated multiple, and a weak business can have an attractive chart for a few months. TI's historical dividend growth and broad catalog are attractive; the current valuation and capex cycle warrant buying only when expected return compensates for uncertainty. We have not used a price target from an analyst consensus as fair value, nor have we extrapolated a moving-average crossover. The neutral-to-positive technical evidence supports staying engaged and waiting for confirmation at support. It does not override the need for a cash-return case that includes incentives, debt, and acquisition costs.

Global parallels reinforce the caution. European analog and automotive suppliers can sustain trends when orders broaden and customers refill inventories, then reverse as OEM production plans change. Indian semiconductor-adjacent industrial suppliers often rally strongly on policy or demand expectations before reported cash generation confirms the thesis. Neither comparison is a signal for TXN; each is a reminder that trend quality and fundamentals need to agree. The current snapshot provides a clean data point: Stage 2, positive moving-average structure, moderate RSI, weak industry breadth, and a no-FOMO support range. Our tactical posture is wait for support or confirmation rather than immediate buy. In the next chapter, we combine this chart evidence with valuation and operational scenarios to reach the final verdict.

Volume should be read as evidence of participation, not as a verdict. The November 6 session had turnover above the snapshot's 30-day average, and the price advanced sharply. That can reflect institutional buying, short covering, index flows, or earnings reaction; the public snapshot cannot distinguish those causes. We should look for repeated accumulation days and relative-strength improvement against the S&P 500 and the semiconductor group. The

sector's 26% above-200-day reading suggests many peers remain below trend, so TXN may be acting as an individual leader. Leadership can persist, but it can also be vulnerable if the group does not confirm. The chart is constructive but not sufficient to establish broad institutional accumulation.

Support is not a fixed wall. The $292.56 nearest-support level and $280.15–$296.95 buy zone are algorithmic references derived from current moving averages and volatility. Price can cross them during normal trading, and earnings gaps can jump over stop levels. The $273.39 invalidation point is therefore a thesis-risk marker, not a guaranteed execution price. If volatility expands, a trader should size risk accordingly and refresh the levels. A long-term investor can use a wider time horizon, but should still be clear about what fundamental evidence would invalidate the thesis. We recommend treating the band as a place to reassess, not a mechanical buy order.

**NARRATOR'S REFLECTION**

Step back from the ticker. Does the price trend support a near-term entry, and what levels would invalidate that reading? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

CHAPTER IX

## Bull vs. Bear, Likely Outcomes & Our Action Plan

> *"It is not necessary to do extraordinary things to get extraordinary results."*
>
> — WARREN BUFFETT

The bull case begins with the product economics. TI sells foundational components across thousands of applications, with long design lives and a strong analog base. Customers value performance, support, and continuity; many do not want to requalify a working component. Internal fabs and 300mm technology can lower cost per chip and improve supply control when utilization rises. The 2026 recovery has spread across industrial, automotive, and data center, and Q3 revenue rose 28% year over year with operating profit rising 73%. Data-center power requirements may add a new source of growth alongside the classic factory and vehicle franchises. TI's direct channel, online tools, and broad catalog allow it to reach both global OEMs and smaller engineers. If current volumes mature into a higher baseline, fab investments could produce better gross margins, strong operating cash, and continued per-share returns. That is the strongest version of the compounding thesis.

The bear case starts with price and fixed costs. At $295.73, TXN carries a trailing P/E above 38 and a price-to-free-cash-flow multiple near 37 on the current snapshot. The shares have already advanced about 84% in a year. Earnings are recovering from a trough but remain below 2022's peak; a slower industrial or automotive cycle could stall the denominator. New fabs require billions of dollars before their output earns mature returns, and 2025 gross margin declined as expansion costs rose. Adjusted free cash flow adds CHIPS incentive receipts, so the headline cash number is not identical to GAAP operating cash less capex. If customers over-order into a recovery, inventory digestion could return. Silicon Labs introduces integration and funding exposure while TI is still financing capacity. Weak semiconductor breadth makes the recent rally more vulnerable to a sector reversal. This is not a claim that TI is broken; it is a claim that a premium price can magnify ordinary cycle risk.

We can steelman both sides without pretending the evidence is symmetrical. The bull side has actual reported revenue and profit acceleration, a product portfolio with genuine longevity, and physical assets already being qualified and ramped. The bear side has equally concrete constraints: a sharp stock re-rating, cycle-sensitive sales, fixed manufacturing costs, a valuation that discounts continued growth, and an acquisition not yet closed. Management's CEO described Q3 as "continued broad growth in industrial, automotive and data center." That description is supported by the overall acceleration, but segment-by-end-market details remain limited and the company's optimistic outlook should not be mistaken for a complete demand ledger. The independent question is which outcome is better compensated at the current quote.

Our probability-weighted view is 25% bull, 50% base, and 25% bear over roughly twelve months. The bull outcome assumes continued broad demand, further margin recovery, effective use of the new fabs, and investor willingness to sustain a quality premium. The base outcome assumes a normal cyclical rebound followed by slower growth, with depreciation and investment limiting the pace of free-cash-flow conversion. The bear outcome assumes that end markets pause, customers draw down stock, fab utilization disappoints, and valuation compresses. These are scenario weights, not a statistical forecast. We do not set a per-share price target because we do not have a fully reconciled long-range model for depreciation, CHIPS incentives, acquisition financing, and post-deal shares. We would rather state the missing bridge than invent precision. The market snapshot's analyst target is not our estimate.

For short-term trading, the verdict is WAIT_FOR_SUPPORT. Stage 2 price structure is constructive and the snapshot's support zone is $280.15–$296.95, but the 52-week rally and weak industry breadth argue against chasing. A tactical plan would require a fresh snapshot, a successful support retest, and a defined risk limit; the published invalidation reference is $273.39. For the medium term, the verdict is WATCH / ACCUMULATE ONLY ON EVIDENCE: next quarterly results should confirm broad demand, industrial recovery, margin leverage, and a credible capital-spending path. For long-term compounding, TI remains a high-quality candidate because of the catalog, customer longevity, and manufacturing model, but at today's multiple we prefer patience and staged entry rather than a full position. Quality is earned; price still matters.

Our no-FOMO plan uses $280.15–$296.95 as a monitor-only support band from the snapshot, with $273.39 as a technical invalidation reference. These are not price targets or guaranteed support. If TXN holds the band, the next gate is evidence: another quarter of organic growth, continued operating margin improvement, reasonable inventory, and separate disclosure of capex and incentive-supported cash flow. If price reclaims the recent resistance and sector breadth improves, a momentum allocation could be reconsidered. If it breaks the stop reference or reported fundamentals deteriorate, we step back and rebuild the thesis. We will not average down solely because the brand is strong, and we will not buy simply because the technical classifier says "BUY_AT_CMP_NO_FOMO." The technical timing gate allows consideration; our valuation judgment caps the action at waiting.

The action plan for current holders differs from the plan for new money. A long-term owner with a low cost basis may reasonably retain a core position while monitoring the fab return and acquisition, because the investment thesis has not broken. A prospective investor must accept today's premium price and cycle risk, so the hurdle is higher. Neither conclusion presumes the user owns TXN; this report is company research, not a personal portfolio audit. Position size should reflect both volatility and the possibility that estimates fall. The annualized dividend yield is meaningful but small relative to the risk of valuation compression. Before any purchase, verify the current quote, current market breadth, updated filings, and updated support levels. The November 6 technical close is the reference for this report, not a standing trading instruction.

The synthesis is therefore a quality business at a demanding, recovery-priced valuation. The moat is real in catalog breadth, manufacturing expertise, channel reach, and design longevity; it is not enough to eliminate end-market cycles or fab-return risk. The near-term chart is constructive enough to keep TXN on the active monitor list. The long-term opportunity remains compelling if free cash flow per share grows after capex normalizes and Silicon Labs contributes without sacrificing capital discipline. At the verified close, those conditions are not yet proven at a price that offers ample protection. Our independent verdict is WAIT_FOR_SUPPORT, conviction tier TIER_1, strategy bucket CORE_ALPHA, with no assigned fundamental target pending a full cash-flow bridge. We prefer a pullback and better evidence over a forecast dressed up as precision.

What would change our mind toward a buy? First, a lower entry price that improves expected return without a corresponding deterioration in business fundamentals. Second, another filing showing continued organic growth across multiple end markets, not just a single data-center pocket. Third, margin improvement alongside transparent capex and incentive reporting. Fourth, evidence that the Silicon Labs transaction will be funded and integrated without impairing TI's FCF-per-share ambition. Fifth, a healthier sector tape that confirms the stock's leadership. What would change our mind toward rejection? A sustained decline below the 30-week trend accompanied by falling estimates, rising inventory, weaker operating cash flow, or a deal that materially stretches the balance sheet. These conditions are measurable; they help prevent the thesis from becoming a story that can explain every price move after the fact.

The probability weights should be revisited after each earnings release rather than defended as a permanent forecast. A bull probability rises if bookings convert to sales without unusual distributor loading, gross margin rises through better factory utilization, and free cash flow improves after excluding one-time incentives. The base case remains the largest because TI's established portfolio supports continuing business, but a premium valuation and high capital needs limit upside if growth simply normalizes. The bear probability rises if customers delay platform production, capacity runs ahead of demand, or acquisition integration distracts management. This is a decision framework, not a statistical model. The strongest discipline is to name the variables that would alter the weights and then change the conclusion when the evidence changes.

The three market-reference levels in the action plan are deliberately not presented as fundamental values. The $318.60 pivot R1 is a nearby resistance reference; $334.03 is the snapshot's 52-week high; and $262.78 is the reported 200-day moving average. The base scenario would require the stock to make progress toward resistance while earnings and cash generation confirm recovery; the bull case would require a clean advance through the prior high with broad sector participation; the bear case could revisit the longer-term average if estimates fall or risk appetite contracts. These levels are sourced technical landmarks, not discounted cash-flow targets, and they can become stale quickly.

Our 25% bull / 50% base / 25% bear weights reflect business quality and valuation together. The base receives the largest weight because the most probable path is neither a perfect manufacturing payoff nor a severe franchise break: TI should continue selling useful products, while margins and cash flow normalize through a mixed cycle. The bull case gets meaningful probability because the latest recovery is broad and the product portfolio has long-lived design positions. The bear case remains material because high expectations, weak sector breadth, fixed costs, and the Silicon Labs transaction can interact. If a future 10-Q shows low inventory risk, steady gross-margin recovery, and successful integration funding, the base can move toward bull. If orders reverse or acquired growth comes at an unattractive price, the bear weight should rise.

**NARRATOR'S REFLECTION**

Step back from the ticker. What is the strongest honest bull case, the strongest honest bear case, and what decision follows from both? The useful answer is the one that survives contact with the next filing, the next customer cycle, and the actual cash-flow statement.

**GLOBAL PARALLEL**

Across US analog suppliers, European automotive chipmakers, and Indian industrial electronics vendors, design qualification can make a product sticky while procurement cycles still make orders volatile. Use the parallel to understand the mechanism, not to assume identical economics.

# Notes, Filings & Primary Sources


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*This Masterclass Companion Book was authored for private study, audiobook narration, and disciplined portfolio execution.*

*Verified against the US Research Ledger • Session Cutoff 2026-11-06*